QUASI-AFFINENESS DESCENDS ALONG AN ARBITRARY FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the descent along the finite locally free morphism; no step is referred to the reader.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-quasi-affine-after-field-extension is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Quasi-affine means quasi-compact and an open subscheme of an affine scheme. Properties, Definition properties-definition-quasi-affine, used at proof lines 2790-2791 to extract quasi-compactness. Supplies that X_K, being quasi-affine, is quasi-compact, so that its image X is quasi-compact.
- Criterion for quasi-separatedness through affine opens. Schemes, Lemma schemes-lemma-characterize-quasi-separated, invoked at proof lines 2796-2797. Supplies quasi-separatedness of X.
- Quasi-affineness descends through a filtered limit. Limits, Lemma limits-lemma-limit-quasi-affine, invoked at proof lines 2803-2804. Supplies the index i for which X_i is already quasi-affine.
- Quasi-affineness is local for the Zariski topology and descends along a finite locally free morphism. Properties, Lemma properties-lemma-quasi-affine-locally-closed and Divisors, Lemma divisors-lemma-push-down-quasi-affine, invoked at proof lines 2809-2813. Supplies that X_{k'} is quasi-affine as a closed subscheme and that quasi-affineness descends from X_{k'} to X.

Source reference. Original theorem 00FV: sources/stacks-chow/tags.txt, source lines 585–585. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BD5: sources/stacks-chow/tags.txt, source lines 14303–14303. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 01Z5: sources/stacks-chow/tags.txt, source lines 2580–2580. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01P6: sources/stacks-chow/tags.txt, source lines 2219–2219. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BCK: sources/stacks-chow/tags.txt, source lines 14283–14283. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KO: sources/stacks-chow/tags.txt, source lines 2057–2057. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. If X_K is quasi-affine for some field extension K/k, then X is quasi-affine.*

Proof. Let K/k be a field extension such that X_K is quasi-affine. The morphism $X_K \to X$ is surjective. Hence X is quasi-compact as the image of a quasi-compact scheme (Properties, Definition 01P6). Since X_K is quasi-separated (as an open subscheme of an affine scheme) we see that X is quasi-separated: If $U, V \subset X$ are affine open, then $(U \cap V)_K = U_K \cap V_K$ is quasi-compact and $(U \cap V)_K \to U \cap V$ is surjective. Thus Schemes, Lemma 01KO applies.

Write $K = \operatorname{colim} A_i$ as the colimit of the subalgebras of K which are of finite type over k. Denote $X_i = X \times_{\operatorname{Spec}(k)} \operatorname{Spec}(A_i)$. Since $X_K = \lim X_i$ we find an i such that X_i is quasi-affine (Limits, Lemma 01Z5; here we use that X is quasi-compact and quasi-separated as just shown). By the Hilbert Nullstellensatz (Algebra, Theorem 00FV) the residue field $k' = A_i/\mathfrak{m}$ is a finite extension of k. Hence $X_{k'} \subset X_i$ is a closed subscheme hence is quasi-affine (Properties, Lemma 0BCK). Since $X_{k'} \to X$ is finite locally free we conclude by Divisors, Lemma 0BD5. $\square$